

Vex Library Documentation

Sub-documentation document to Vex Documentation



This document describes how Vex libraries can be created, used, and published to the Vex Library Collection (VexLibC).

Creating A Library

To create a library for Vex, all you must do is create an executable that Vex will run, and follow these instructions below.

Reference point for instructions below:

```
library.opcode.opcode(variables)
```

First, your executable must be named what you want your library to be called. There can be no file extension. Vex will run the command like so:

```
$ ./path/to/library opcode opcode variable1=10
```

Your executable must incorporate reading options separated by spaces. It runs like a normal command would in the terminal, with your options separated by spaces. Your code will take control of what is shown in the terminal until the command is complete, or it can return a value based on output. For example, here is a Vex library called fetch:

```
globalvar currentSecond = fetch.http(url =  
"https://example.com") # This returns the fetched URL
```

Under the hood, that would actually run:

```
$ ./path/to/fetch http url="https://example.com"
```

Your library will then respond with the output, and it will pass the output to the variable, as seen in the code.

Using Event Variables

If you are creating a library, and it runs a daemon, that daemon can interact with variables that the Vex code can see (event:library.var). The code below works in Python:

```
import json
import os

EVENTS_FILE = os.path.join(LIB_DIR, "events.json")

def set_event(event_name, state):
    os.makedirs(LIB_DIR, exist_ok=True)
    data = {}
    if os.path.exists(EVENTS_FILE):
        try:
            with open(EVENTS_FILE, 'r') as f:
                data = json.load(f)
        except Exception:
            pass
    data[event_name] = state
    with open(EVENTS_FILE, 'w') as f:
        json.dump(data, f)
```

Tip: The best way to create and compile a library is by using C++ or Python (Python must be with shebang).

Publishing a Library

To publish your library, the source code before compiling must be provided. Otherwise, your library will not be added to VexLibC. More rules are on the Google Form for uploading libraries.

Form: <https://forms.gle/1hJbDhT1XWqCTLyW6>